Capstone Microphone Requirements

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October 2026

1 Purpose

The purpose of this document is to define the requirements for a microphone that can capture the full range of fundamental frequencies produced by a cello in real time, show why the CB 100 is a good choice.

2 Frequencies

In twelve-tone equal temperament, with A4 tuned to 440 Hz, the fundamental frequency of a note with MIDI number m is

$$f = 440 \cdot 2^{(m-69)/12} \text{ Hz}. \tag{1}$$

Table 1 lists the lowest and highest fundamental frequencies the system is required to capture. The lower bound is the open C string; the upper bound is A5, the top of the range expected from beginner to intermediate players.

Table 1: Cello fundamental frequency range

Bound	Note	MIDI	Frequency (Hz)
Minimum	C2	36	65.41
Maximum	A5	81	880.00

3 Interface

4 Microphone

The Behringer CB 100 was selected due to its low cost, compact size, and ability to connect directly to the Focusrite Scarlett Solo through a standard XLR connection using the interface's 48 V phantom power. Its key specifications are summarized in Table 2.

Table 2: Behringer CB 100 specifications

Parameter	Value
Type	Condenser, gooseneck
Frequency response	50 Hz – 18 kHz
Sensitivity	−50 dBV ±3 dB at 1 kHz
Impedance	200 Ω
Power	48 V phantom
Connector	XLR 3-pin

4.1 Frequency Gain Response

Figure 1 shows the frequency response of the CB 100 provided by the manufacturer. Output is expressed in dBV (decibels relative to 1 V RMS) for a constant input sound pressure.

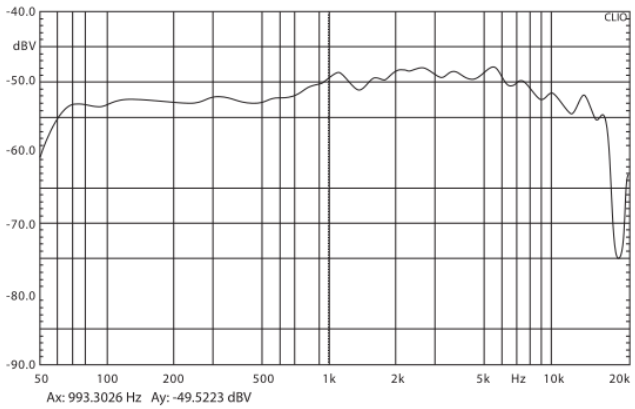


Figure 1: CB 100 frequency response.

The response is approximately flat at −54 dBV from roughly 70 Hz to 500 Hz. At the lowest cello fundamental, C2 (65.41 Hz), the output is within approximately 1–2 dB of this level, satisfying the requirement of less than 3 dB attenuation at 60 Hz. The response remains usable down to 50 Hz, where it is approximately 7 dB below the flat region. Across the full range of fundamentals in Table 1 (65.41–880 Hz) the response varies by only a few decibels. The CB 100 therefore meets the frequency response requirements.